

Problem 1:-

Square upper triangular matrix is invertible iff its main diagonal elements are non-zero.

This is because if determinant is 0 then it will not be invertible. By theorem, determinant of a triangular matrix is the product of diagonal elements.

Problem 2:-

Square lower triangular is invertible iff its main diagonal is non-zero.
Same reason as problem 1.

Problem 3:-

- Onto as well $\because$ finite same dimension transform
- $\text{Ker}(A) = \{0\}$ $\because AX=0 \Rightarrow X=0$
- $\text{Rank}(A) = n$ $\because$ Rank-Nullity theorem
- A invertible $\because$ Bijectivity.
- $\det(A) \neq 0$ $\because$ Invertibility.

Problem 4:-

1. $\det(A) = 1$, A is invertible therefore it is both one-to-one and onto.

2. Since the matrix is not invertible therefore it is not one-to-one and onto since there is no bijectivity. & onto is possible if it is one-to-one in same dimensions and transformations.

3. $\text{rank}(A) = 2$
 $\text{rank}(A) + \text{nullity}(A) = 2$
 $2 + 0 = 2$
 $\text{nullity}(A) = 0$
 $\text{Ker}(A) = 0$

One to one
Not onto as dimensions are changed.

4. $\text{rank}(A) = 2$ because 2 non-zero rows.
 $\text{rank}(A) + \text{nullity} = 3$
 $2 + 1 = 3$
 $\text{Ker}(A) = 1$
Not one-to-one.

$\text{rank}(A) = 2 = \text{number of rows}$

Onto.